WORKED SOLUTION

STEP III 1990, Question 11

Mechanics • Rigid Body Rotation

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Read the question before you go on. It is in the STEP Support Programme's question database:

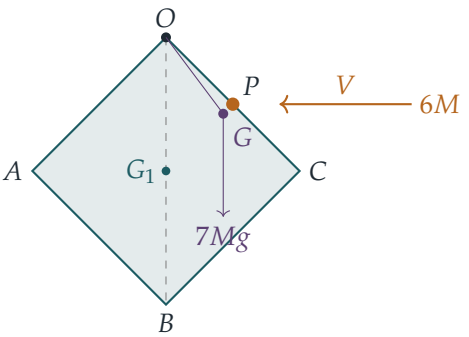
step.maths.org/questions/90-s3-q11

The key idea

The impact is sudden, and the axis at O exerts a large impulsive force during it, so linear momentum is *not* conserved. But that impulse acts at O and so has no moment about O: angular momentum about the axis is conserved through the collision. After that the system is a rigid body on a fixed axis, gravity is the only force doing work, and conservation of energy tells us whether it gets over the top.

Solution

We model the lamina as a uniform rigid square, the particle as a point mass that sticks to the lamina at the instant of impact, and the axis as smooth. Take O as origin. With OB vertically downwards, the centre G_1 of the square is at depth $\sqrt{2}\,a$ (half the diagonal $2\sqrt{2}\,a$), and A and C are at horizontal distance $\sqrt{2}\,a$ either side of OB, also at depth $\sqrt{2}\,a$. The midpoint P of OC is therefore at horizontal distance $a/\sqrt{2}$ from O and depth $a/\sqrt{2}$, with $OP = a$.



(The diagram shows the particle arriving from the right; arriving from the left gives the same answer, as we shall see.)

Moment of inertia about the axis

For a uniform square lamina of side $2a$, the moment of inertia about the perpendicular axis through its centre is $\frac{1}{12}M((2a)^2 + (2a)^2) = \frac{2}{3}Ma^2$ (by the perpendicular axis theorem, adding the two in-plane axes, each $\frac{1}{3}Ma^2$). By the parallel axis theorem, with $OG_1 = \sqrt{2}\,a$,

$$I_{\text{lamina}} = \tfrac{2}{3}Ma^2 + M(\sqrt{2}\,a)^2 = \tfrac{8}{3}Ma^2.$$

The particle ends up at P, a distance a from O, so the combined body has

$$I = \tfrac{8}{3}Ma^2 + 6Ma^2 = \tfrac{26}{3}Ma^2.$$

The impact

> **Ask yourself** Why can't we use conservation of linear momentum for the collision?

Because the axis is fixed: to stop O moving, it must deliver an impulse to the lamina during the impact, and that impulse is unknown. Its line of action passes through O, though, so it has zero moment about O, and gravity gives no impulse in an instant. So angular momentum about the axis is conserved.

Just before impact, the particle's angular momentum about O is its momentum $6MV$ times the perpendicular distance from O to its (horizontal) line of motion, which is the depth of P, namely $a/\sqrt{2}$. The lamina is at rest. Just afterwards the whole body turns with angular speed ω, so

$$\tfrac{26}{3}Ma^2\,\omega = 6MV \cdot \frac{a}{\sqrt{2}} \qquad \Longrightarrow \qquad \omega = \frac{9V}{13\sqrt{2}\,a}.$$

The kinetic energy immediately after impact is then

$$\tfrac{1}{2}I\omega^2 = \tfrac{1}{2} \cdot \tfrac{26}{3}Ma^2 \cdot \frac{81V^2}{338a^2} = \frac{27}{26}MV^2.$$

Making complete revolutions

The combined centre of mass G (mass $7M$) has horizontal displacement and depth from O

$$\frac{M \cdot 0 + 6M \cdot \frac{a}{\sqrt{2}}}{7M} = \frac{3\sqrt{2}}{7}a, \qquad \frac{M \cdot \sqrt{2}\,a + 6M \cdot \frac{a}{\sqrt{2}}}{7M} = \frac{4\sqrt{2}}{7}a,$$

so $OG = \frac{\sqrt{2}}{7}a\sqrt{3^2 + 4^2} = \frac{5\sqrt{2}}{7}a$.

The body is rigid and pinned at O, so (unlike a particle on a string) it cannot "fall off" its circle; it makes complete revolutions exactly when it still has positive kinetic energy at the position of highest potential energy, which is when G is vertically above O. From its starting depth $\frac{4\sqrt{2}}{7}a$ below O to a height $\frac{5\sqrt{2}}{7}a$ above O, G rises

$$\frac{4\sqrt{2}}{7}a + \frac{5\sqrt{2}}{7}a = \frac{9\sqrt{2}}{7}a,$$

whichever way the body turns, so the direction from which the particle came doesn't matter. Only gravity does work after the impact (the smooth axis does none), so by conservation of energy we need

$$\frac{27}{26}MV^2 > 7Mg \cdot \frac{9\sqrt{2}}{7}a = 9\sqrt{2}\,Mga,$$

that is,

$$\boxed{V^2 > \frac{26\sqrt{2}}{3}\,ga.}$$

The mass M cancels, as it must, since every mass in the problem is a multiple of M. (At exactly equality the body would only creep towards the top position, never passing it, so the inequality is strict.)

Checks

- The moment of inertia, centre of mass (tan of its angle from the downward vertical is $\frac{3}{4}$) and kinetic energy factor $\frac{27}{26}$ were all confirmed numerically, the first by integrating over a grid on the tilted square.
- Stepping the equation of motion $I\ddot{\theta} = -7Mg\,x_G$ numerically with $a = 1.3$, $g = 9.81$: with V^2 at 98% of $\frac{26\sqrt{2}}{3}ga \approx 15.6$ the body swings back, while at 102% it goes right round, for either direction of approach.
- Sanity check on the energy: only $\frac{27}{26}MV^2$ out of the particle's $3MV^2$ survives the impact, as expected for a perfectly inelastic collision.

Common slips

- Conserving linear momentum through the impact: the axis gives an impulse, so it isn't conserved.
- Using the distance $OP = a$ as the moment arm of the particle's momentum, instead of the perpendicular distance $a/\sqrt{2}$ to its line of motion.
- Forgetting that the centre of mass has moved off OB once the particle sticks, so the rise to the top is not simply twice the original depth.

Beyond STEP

This is the mechanics of rigid bodies rotating about a fixed axis, where impulsive reactions at the axis make angular momentum, rather than linear momentum, the conserved quantity in collisions. A neat follow-up is the *centre of percussion*: the point where a blow produces no impulsive reaction at the axis at all, which is why a cricket bat has a "sweet spot". Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and angular impulse carefully.

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